BACHELOR WITH APPLIED COMPUTING AS MINOR
SIXTH SEMESTER
(FOR STUDENTS WITH MAJOR IN COMPUTER APPLICATIONS / INFORMATION TECHNOLOGY)

ACP622N: APPLIED COMPUTING _ ARTIFICIAL INTELLIGENCE

CREDITS: THEORY (3) PRACTICAL (1)

MAX. MARKS: 100 MIN. MARKS: 36

COURSE LEARNING OUTCOMES:

1. *Study the concepts of Artificial Intelligence.*
2. *Learn the methods of solving problems using Artificial Intelligence.*
3. *Learn the knowledge representation techniques, reasoning techniques and planning.*

THEORY (3 CREDITS)

PREREQUISITE: Data Structures and Analysis of Algorithm

UNIT 1: AI AND TURING TEST: (15 LECTURES)

Introduction to Artificial Intelligence, Background and Applications, Turing Test, Rational Agent approaches to AI, Introduction to Intelligent Agents, structure, behaviour and environment.

UNIT 2: SEARCHING PROBLEM SPACE: (15 LECTURES)

Problem Characteristics, Production Systems, Informed vs Uninformed Strategies, Hill Climbing and its Variations, Heuristics Search Techniques: Best First Search, A* algorithm.
The Wumpus World Environment, Acting and reasoning in the Wumpus world, Representation, Reasoning, Logic, Prepositional Logic and First-Order Logic, Rules of Inferencing: Resolution Principle.

UNIT 3: UNCERTAIN KNOWLEDGE AND REASONING: (15 LECTURES)

Acting under Uncertainty, handling uncertain knowledge, Uncertainty and rational decisions, Basic Probability Notation, Prior probability, Conditional probability, joint probability distribution, Bayes' Rule and Its Use, Applying Bayes' rule: The simple case, Normalization, Dampster's Shafer's Theorem.

TUTORIAL (1 CREDIT):

1. Given a real life situation
   - identify the characteristics of the environment
   - identify the percepts available to the agent
   - identify the actions that the agent can execute
   - suggest the performance measures to evaluate the agent
   - recommend the architecture of the desired intelligent agent
2. Undertake a comparative study on popular programming languages used in AI.
3. Translate commonly used English sentences into their equivalent logical expressions.
4. Given a problem description, formulate it in terms of a state space search problem. Analyze the given problem and identify the most suitable search strategy for the problem. Make an analysis of the properties of proposed algorithms in terms of
   - time complexity
   - space complexity
   - optimality
5. Write a Program to Implement Breadth First Search.
6. Write a Program to Implement Depth First Search.
7. Write a program to implement Hill Climbing Algorithm
8. Write a program to implement A* Algorithm.
9. Write a program to implement Tic-Tac-Toe game.
10. Write a Program to Implement 8-Puzzle problem.
11. Write a Program to Implement Water-Jug problem.
12. Write a Program to Implement Alpha-Beta Pruning using Python.
13. Write a Program to Implement 8-Queens Problem.
14. Build a Fuzzy inference system for the Tipping Problem.
    Given two sets of numbers between 0 and 5 (where 0 is for very poor, and 5 for excellent) that respectively represent quality of service and quality of food at restaurant, what should tip be?
15. Solve 2-input 1-output project risk prediction problem using Mamdani Inference. Make necessary assumptions.

TEXT BOOK:

1. DAN.W. Patterson, Introduction to A.I and Expert Systems - PHI, 2007.

REFERENCES:

1. Russell & Norvig. Artificial Intelligence-A Modern Approach, LPE, Pearson Prentice Hall, 2nd edition. 2005.
2. Ivan Bratko,"Prolog Programming for Artificial Intelligence", 3 rd Edition, Pearson Education.
3. Rich & Knight, Artificial Intelligence - Tata McGraw Hill. 2nd edition, 1991.
4. W.F. Clocksin and Mellish, Programming in PROLOG, Narosa Publishing House, 3rd edition, 2001.